

Bachelor of Technology — Microelectronics and VLSI

Table 1: Microelectronics and VLSI — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	54
	Discipline Electives	12
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Microelectronics and VLSI — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-212	Digital System Design	3-0-2-4	4
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-302	Control Systems	3-0-2-4	4
5	VL-201	Semiconductor Device for ICs	3-0-0-3	3
6	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
7	EE-211	Analog Circuit Design	2-0-2-3	4
8	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
9	VL-311	CMOS Processing and Practicum	3-0-2-4	4
10	VL-312	Electronic System Packaging	3-0-0-3	3
11	VL-326	Computer Organization	3-0-2-4	4
12	VL-401	RTL Design and Verification	2-0-2-3	3
13	VL-402	RF IC Design	3-0-0-3	3
14	VL-403	CMOS Digital IC Design	3-0-2-4	4
15	VL-404	CMOS Analog IC Design	3-0-2-4	4
16	VL-405	Design for Testability	3-0-2-4	4
Total				54

Table 3: B. Tech Microelectronics and VLSI — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Microelectronics and VLSI — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Microelectronics and VLSI — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-212	Digital System Design	3-0-2-4	4
6	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
7	DC	EE-302	Control Systems	3-0-2-4	4
8	DC	VL-201	Semiconductor Device for ICs	3-0-0-3	3
Total					25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Microelectronics and VLSI — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	4
3	DC	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
4	DC	VL-311	CMOS Processing and Practicum	3-0-2-4	4
5	DC	VL-312	Electronic System Packaging	3-0-0-3	3
6	DC	VL-326	Computer Organization	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Microelectronics and VLSI — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	VL-401	RTL Design and Verification	2-0-2-3	3
2	DC	VL-402	RF IC Design	3-0-0-3	3
3	DC	VL-403	CMOS Digital IC Design	3-0-2-4	4
4	DC	VL-404	CMOS Analog IC Design	3-0-2-4	4
5	DC	VL-405	Design for Testability	3-0-2-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Microelectronics and VLSI — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Microelectronics and VLSI — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	VL-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Microelectronics and VLSI — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	VL-010	Internship		2
2	MTP	VL-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 12 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Microelectronics and VLSI — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	EE-210P	Digital Systems Design Lab	1
2	EE-507	Transmission Lines and Basic Microwave Engineering	4
3	EE-516	Biomedical Systems	4
4	EE-520	Microelectronic Devices and Modelling	3
5	EE-523	Digital VLSI Architecture Design	4
6	EE-529	Embedded Systems	4
7	EE-554	Low Power VLSI Design	3
8	EE-574	Biomedical Signal and Image Analysis	3
9	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
10	EE-611	VLSI Technology	3

Microelectronics and VLSI — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-615	Nanoelectronics and Micronanofabrication	3
12	EE-619	Mixed Signal VLSI Design	4
13	EE-621	Radiating systems	4
14	EE-622	Microwave Integrated Circuits	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	VL-498P	Major Technical Project - I (compulsory; counted as DE)	4
16	VL-499P	Major Technical Project - II (compulsory; counted as DE)	4